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POWER CONVERTER WITH DISPLAY

Abstract

A power converter may comprise converter circuitry, a housing and a display. The converter circuitry is configured to convert electrical energy. The housing may comprise a mounting groove. The display may comprise a display screen provided in the mounting groove and a protective layer covering the mounting groove. The protective layer may have a translucent area and an opaque area adjacent to the translucent area. The display screen may comprise a display area located corresponding to the translucent area, and a projection of the translucent area toward the display screen may cover the display area.

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATION

[0001] The present application claims priority to CN 202420369918.2, filed on Feb. 27, 2024, which is incorporated by reference by its entirety.

FIELD

[0002] The present disclosure relates to the technical field of energy storage systems, in particular to a converter (e.g., power converter).

BACKGROUND

[0003] With the development of science and technology, energy storage systems and power generation systems are constantly updated and developed, bringing convenience to people's lives. For example, domestic solar power generation systems can provide electricity for family life. As another example, mobile energy storage systems can facilitate people to use a variety of electrical appliances outdoors.

[0004] In a working process of the energy storage system and power generation system, converters are often needed to convert current. For example, electronic devices such as converters are needed to convert between direct current (DC) and alternating current (AC). A converter can be equipped with a display screen to display various kinds of work-related information, but the current display effect of the display screen needs to be improved.

SUMMARY

[0005] A technical problem mainly solved by the present disclosure is to provide a converter with an improved display effect of a display area of a display screen.

[0006] In order to solve the technical problem, a technical solution adopted by the disclosure is as follows. A power converter with an improved display is provided. For example, a power converter may comprise converter circuitry, a housing and a display. The converter circuitry is configured to convert electrical energy. The housing may comprise a mounting groove. The display may comprise a display screen provided in the mounting groove and a protective layer covering the mounting groove. The protective layer may have a translucent area and an opaque area adjacent to the translucent area. The display screen may comprise a display area located corresponding to the translucent area, and a projection of the translucent area toward the display screen may cover the display area

[0007] Beneficial effects of the disclosure are: different from the power converters in the related art, a part of the protective layer used with the display screen is set to be translucent, and the other part thereof is set to be opaque, thereby forming a translucent area and an opaque shielding area. The translucent area can allow the lighted characters and images on the display area to be seen by one or more users, and the translucent area can also shield the unlighted characters and images on the display area, thus limiting visual interference caused by the unlighted characters and images and making the lighted characters and images stand out, thereby improving the display effect of the display area and facilitating the users to clearly observe the display information on the display area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a three-dimensional structure diagram of a converter according to an example of the present disclosure;

[0009] FIG. **2** is a front view of the converter according to an example of the disclosure;

[0010] FIG. **3** is a three-dimensional structure diagram of the converter whose surface cover and display module are not assembled according to an example of the present disclosure;

[0011] FIG. **4** is a cross-sectional structure diagram cut along a line D-D of the converter shown in FIG. **2** with part of components hidden;
[0012] FIG. **5** is a schematic diagram showing an assembly process of the display module shown in FIG. **3**;
[0013] FIG. **6** is a structure diagram of a protective member of the converter according to an example of the disclosure;
[0014] FIG. **7** is a structure diagram of a display screen assembly of the converter according to an example of the disclosure;
[0015] FIG. **8** is another cross-sectional structure diagram of the converter according to an example of the disclosure with part of components hidden;
[0016] FIG. **9** is a structure diagram of part A shown in FIG. **4**;
[0017] FIG. **10** is a structure diagram of part B shown in FIG. **9**; and
[0018] FIG. **11** is a structure diagram in which a light source assembly and a radar assembly are mounted on the surface cover.

DETAILED DESCRIPTION

[0019] Technical solutions in the present disclosure will be clearly and fully described with reference to the accompanying drawings. Obviously, the examples to be described are merely part of the disclosure, but not all examples of the disclosure. Based on the examples of the disclosure, all other examples obtained by those of ordinary skill in the art without inventive work shall fall within the scope of the disclosure.
[0020] In a working process of the energy storage system and power generation system, electronic devices such as converters are often needed to convert electrical energy such as current. For example, converters are needed to convert between DC and AC. The converter can be equipped with a display screen to display various kinds of work-related information, but the current display effect of the display screen of the converter to display various kinds of work-related information needs to be improved. For example, internal components or structures of the converter corresponding to a non-display area can be seen. In order to solve the problem, the present disclosure provides the following examples.
[0021] As shown in FIGS. **1** to **4**, a converter **1** (e.g., a power converter) according to an example of the present disclosure includes a converter body **100** and a display module **200**. The converter body **100** is configured to convert current, and a mounting groove **111** is provided on an outer side of the converter body **100**. As shown in FIG. **3**, the display module **200** includes a display screen assembly **210** and a protective member **220** (e.g., a protective layer, a glass layer, a plastic or resin cover, a metal frame, a screen protector). The display screen assembly **210** can be provided in the mounting groove **111**, and the protective member **220** can be provided to cover the mounting groove **111**.
[0022] The converter **1** may be an energy storage converter, an energy storage inverter, a power converter, or the like. For example, the converter body **100** can be configured to convert DC to DC, AC to AC, AC to DC, and DC to AC. The converter body **100** may comprise one or more circuits (e.g., circuitry) and the circuits may comprise transformers. The converter body **100** may be connected with an energy storage system, so as to manage a charging and discharging process of the energy storage system. The converter body **100** may be connected with one or more electrical equipment, so as to provide matching current for the electrical equipment. The converter body **100** itself may also have an energy storage unit (e.g., battery).
[0023] The protective member **220** may include a translucent area **221** and an opaque shielding area **222** which are adjacently to each other. The display screen assembly **210** is provided with a display area **211** which can be located corresponding to the translucent area **221**, and a projection of the translucent area **221** toward the display screen assembly **210** may cover the display area **211**.
[0024] In some examples, the translucent area **221** and the opaque shielding area **222** may present the same color. For example, both the translucent area **221** and the opaque shielding area **222** may

be in black.
[0025] Further, the opaque shielding area **222** may be connected to an edge of the translucent area **221**. For example, the opaque shielding area **222** may be adjacently provided around the edge of the translucent area **221**.
[0026] The display screen assembly **210** may be configured to present a relevant working state of the converter **1** to one or more users. For example, the display area **211** can display working parameters of the converter **1**, the electric quantity of the energy storage system or energy storage unit, a network connection state, and/or the like. The protective member **220** is located at a side of the display screen assembly **210** facing the outside, so as to protect the display screen assembly **210** and other components in the mounting groove **111** from mechanical damage, stain pollution, and other damage from the outside.
[0027] The translucent area **221** allows the light emitted from the display area **211** to pass through, so that a user can see the lighted characters displayed in the display area **211**. The display area **211** is provided opposite to (e.g., corresponding to) the translucent area **221**, and a projection of the translucent area **221** toward the display screen assembly **210** covers the display area **211** (e.g., a size of the translucent area **221** is the same as or larger than a size of the display screen assembly **210**), which is conducive to clearly observing the information displayed in the display area **211** (e.g., when the user horizontally looks at the display area **211**). The opaque shielding area **222** can shield an area irrelevant to the display information inside the mounting groove **111** to highlight the display effect of the display information and beautify the appearance of the converter **1**. In addition, providing the opaque shielding area **222** makes the appearance of the converter **1** more elegant.
[0028] The display screen assembly **210** displays information by lighting characters, images and the like on the display area **211**. By setting the translucent area **221** to be translucent, the translucent area **221** can allow the lighted characters and images on the display area **211** to be seen by users, and the translucent area **221** can also shield the unlighted characters and images on the display area **211**, thus limiting visual interference caused by the unlighted characters and images and making the lighted characters and images stand out, thus improving the display effect of the display area **211** and facilitating the users to clearly observe the display information on the display area **211**.
[0029] In some examples, as shown in FIG. **5**, the display screen assembly **210** includes a display screen **214** and a frame **213** surrounding the display screen **214**. The frame **213** can be used to support and fix the display screen **214**, the display area **211** is provided on the display screen **214**, and the opaque shielding area **222** can be used to shield the frame **213** for aesthetic effect.
[0030] In some examples, the display screen **214** is provided with a blank space around the display area **211**, and the opaque shielding area **222** can be used to shield an outer edge of the blank space for aesthetic effect.
[0031] As shown in FIGS. **4** to **6**, the protective member **220** includes a transparent substrate **223**. The protective member **220** further includes a light shielding layer **224** covering an inner side of the transparent substrate **223** facing the mounting groove **111**, and an area where the light shielding layer **224** covers the transparent substrate **223** forms the opaque shielding area **222**.
[0032] By providing the light shielding layer **224**, it is convenient for the protective member **220** to realize a light shielding function, and it is beneficial to simplify a production process of the protective member **220**. For example, the light shielding layer **224** is an opaque black ink layer, and in the production process of the protective member **220**, an area corresponding to the transparent substrate **223** can be coated with opaque black ink to form the opaque shielding area **222**. As another example, the light shielding layer **224** is an opaque black film, and in the production process of the protective member **220**, the opaque black film can be attached to an area corresponding to the transparent substrate **223** to form the opaque shielding area **222**. In other examples, the light shielding layer **224** is black chemical plating. In other examples, the protective

member **220** includes an opaque substrate that can be used as the opaque shielding area **222**.

[0033] As shown in FIGS. **4** to **6**, the protective member **220** further includes a translucent layer **225** covering an inner side of the transparent substrate **223** facing the mounting groove **111**, visible light transmittance of the translucent layer **225** may be 5% to 25%, and an area where the translucent layer **225** covers the transparent substrate **223** forms the translucent area **221**.

[0034] The light shielding layer is made of opaque material, and the translucent layer is made of translucent material. For example, the opaque material includes at least one of opaque black ink, opaque black film and black chemical plating; the translucent material includes at least one of translucent black ink or translucent film.

[0035] By providing the translucent layer **225**, it is convenient for the translucent area **221** to achieve a translucent effect, and it is beneficial to simplify the production process of the protective member **220**. For example, the translucent layer **225** is a translucent ink layer, and in the production process of the protective member **220**, an area corresponding to the transparent substrate **223** can be coated with translucent black ink to form the translucent area **221**. As another example, the translucent layer **225** is a translucent film, and in the production process of the protective member **220**, the translucent film can be attached to an area corresponding to the transparent substrate **223** to form the translucent area **221**. In other examples, the protective member **220** includes a translucent substrate that can be used as the translucent area **221**.

[0036] The light shielding layer **224** may be provided around the translucent layer **225**. In some examples, the transparent substrate **223** may be coated with opaque black ink to form the opaque shielding area **222**, and the transparent substrate **223** may be coated with translucent black ink to form the translucent area **221**, in which the opaque black ink is applied around the translucent black ink.

[0037] As shown in FIG. **7**, the display screen assembly **210** includes a translucent film **212** attached to the display area **211**, and visible light transmittance of the translucent film **212** may be 30% to 60%.

[0038] By providing the translucent film **212**, it is possible to allow the lighted characters and images on the display area **211** to be seen by users and shield the unlighted characters and images on the display area **211**, thereby improving the display effect of the display area **211**. By attaching the translucent film **212** to the display area **211**, it is beneficial to simplify the production process of the display screen assembly **210**.

[0039] The color of the translucent film **212** may be black or silver gray. Further, the color of the translucent film **212** may be consistent with the color of the translucent layer **225**.

[0040] The transparent substrate **223** is tempered glass. Thus, the protective member **220** can have good mechanical strength, chemical stability and service life, the protective effect of the protective member **220** on the display screen assembly **210** can be improved, and the protective member **220** itself is not easy to be scratched, age and discolor, which is beneficial to clearly display the display information on the display area **211** to users.

[0041] A thickness of the protective member **220** may range from 2 mm to 4 mm. For example, the thickness of the protective member **220** is 3 mm. Thus, the protective member **220** can have good mechanical strength and be light and thin.

[0042] As shown in FIGS. **4** to **6**, an outer side of the protective member **220** facing away from the mounting groove **111** may be provided with at least one of an anti-fingerprint coating **226** and an anti-glare coating **227**.

[0043] The anti-fingerprint coating **226** has small surface energy, a large contact angle, and good antifouling and waterproof performance, so that the protective member **220** can be restricted from being contaminated with a fingerprint, and thus the surface of the protective member **220** can be kept clean, which is beneficial to clearly display the display information on the display area **211** to the users. Specifically, the anti-fingerprint coating **226** can be formed on the transparent substrate **223** by AF (Anti-fingerprint) treatment.

[0044] The anti-glare coating **227** can have an effect of multi-angle diffuse reflection, thereby increasing a viewing angle of the display area **211** and reducing a problem that the protective member **220** reflects external light, which is beneficial to clearly display the display information on the display area **211** to the users. Specifically, the anti-glare coating **227** can be formed on the transparent substrate **223** by AG (Anti-glare) treatment.
[0045] In some examples, the outer side of the protective member **220** facing away from the mounting groove **111** is provided with the anti-fingerprint coating **226** and the anti-glare coating **227** simultaneously. In the production process of the protective member **220**, the protective member **220** can be AG-treated to obtain the anti-glare coating **227**, and then the protective member **220** can be AF-treated to obtain the anti-fingerprint coating, so that the anti-fingerprint coating covers the anti-glare coating **227**.
[0046] In some examples, the protective member **220** can also form an anti-reflection coating by anti-reflection (AR) treatment to reduce the problem that the protective member **220** reflects external light.
[0047] As shown in FIGS. **3**, **5**, and **8**, the display module **200** may further include a backing plate member **230** provided on the converter body **100**, and the protective member **220** and the display screen assembly **210** are respectively provided on two opposite sides of the backing plate member **230**. The backing plate member **230** is provided with a mounting hole **231**, and the display area **211** is exposed to the translucent area **221** through the mounting hole **231**.
[0048] By providing the backing plate member **230**, it is convenient to connect the display screen assembly **210** with the protective member **220**, and thus control a distance L**1** between the display screen assembly **210** and the protective member **220** within a certain range, which is beneficial to clearly display the display information on the display area **211** to users.
[0049] The mounting hole **231** may penetrate the backing plate member **230** and be opposite to the translucent area **221**. In some examples, the display screen assembly **210** is located outside the mounting hole **231**, and the display area **211** is opposite to the mounting hole **231**. In other examples, the display screen assembly **210** extends from the outside of the mounting hole **231** to the inside of the mounting hole **231**, and the display area **211** is located outside and opposite to the mounting hole **231** or inside the mounting hole **231**, which is beneficial to reduce the distance L**1** between the display screen assembly **210** and the protective member **220**, and thus clearly display the display information on the display area **211** to the users.
[0050] The backing plate member **230** is connected and fixed to the protective member **220** by sticking. Specifically, the backing plate member **230** may be connected and fixed to the opaque shielding area **222** by sticking.
[0051] As shown in FIG. **8**, the converter body **100** includes a surface cover **110** and a main housing **120**, the mounting groove **111** is formed on the surface cover **110**, and a side of the surface cover **110** facing away from the mounting groove **111** is connected with the main housing **120**. The backing plate member **230** is detachably mounted on the surface cover **110** through a fixing member **130**. The main housing **120** is provided with an avoidance hole **121** and a cover plate **122** detachably covering the avoidance hole **121**. The avoidance hole **121** penetrates the main housing **120** and is opposite to the fixing member **130**, and the fixing member **130** is exposed to the outside through the avoidance hole **121** after the cover plate **122** is detached.
[0052] The fixing member **130** passes through the surface cover **110**. When the display module **200** needs to be detached or replaced, the cover plate **122** can be opened to expose the fixing member **130** through the avoidance hole **121**, and then the fixing member **130** can be detached from the backing plate member **230**, so that the display module **200** and the surface cover **110** are disconnected, and the display module **200** can be detached. Thus, the display module **200** can be detached without detaching the entire main housing **120** from the surface cover **110**, and it is convenient to replace and maintain the display module **200**.
[0053] Specifically, the surface cover **110** and the main housing **120** can enclose an

accommodating cavity **123** which can accommodate various electronic devices of the converter **1** for converting current. The avoidance hole **121** communicates with the accommodating cavity **123**. The fixing member **130** is a screw, the surface cover **110** is provided with a first connecting hole **114** therethrough, and a side of the backing plate member **230** facing the surface cover **110** is provided with a first threaded hole **233**. The fixing member **130** passes through the first connecting hole **114** and extends into the first threaded hole **233** to connect the display module **200** and the surface cover **110**.
[0054] Further, as shown in FIGS. **5** and **8**, the backing plate member **230** includes a plate body **232** and a first connecting part **234** which protrudes from a side of the plate body **232** facing the surface cover **110**, the first threaded hole **233** is provided at the first connecting part **234**, and the display screen assembly **210** and the protective member **220** can be connected to two sides of the plate body **232**. Thus, a thickness of the plate body **232** can be reduced, and occupied space of the backing plate member **230** can be reduced, which is beneficial to reduce the distance L**1** between the display screen assembly **210** and the protective member **220**.
[0055] There may be a plurality of first connecting parts **234** which are respectively provided at two ends of the plate body **232** along its own length direction. A plurality of first connecting holes **114** correspond to the first connecting parts **234** respectively.
[0056] As shown in FIGS. **4**, **9** and **10**, the display screen assembly **210** may be provided with a frame **213** which is provided around the display area **211**. The display module **200** further includes a sealing member **240** (e.g., foam gasket, adhesive seal, or silicone layer) which abuts between the frame **213** and the protective member **220**. Specifically, the sealing member **240** abuts between the frame **213** and the opaque shielding area **222**. Alternatively, the sealing member **240** abuts between the frame **213** and the backing plate member **230**.
[0057] By providing the frame **213**, pressure on the display area **211** can be reduced, which is beneficial for the display area **211** to present a good display effect. The sealing member **240** can limit water leakage and light leakage between the display area **211** and the protective member **220**, which is beneficial to improve the display effect of the display area **211**.
[0058] Further, the sealing member **240** can be made of flexible material, which can play a buffering role, reduce mechanical impact on the display area **211** during an assembly process or during mounting and use in application scenarios, and facilitate assembly of the display screen assembly **210** in place.
[0059] Further, as shown in FIG. **3** to FIG. **5**, the backing plate member **230** includes a plate body **232** and a second connecting part **235** which protrudes from the side of the plate body **232** facing the surface cover **110** and is provided with a second threaded hole **236**. The frame **213** is provided with a second connecting hole **215** therethrough, and a screw passes through the second connecting hole **215** and extends into the second threaded hole **236**, so that the frame **213** and the backing plate member **230** are connected.
[0060] There may be a plurality of second connecting parts **235** which are distributed at intervals around the mounting hole **231**. The second connecting holes **215** correspond to the second connecting parts **235**, respectively.
[0061] As shown in FIGS. **4**, **9** and **10**, the distance L**1** between the display screen assembly **210** and the protective member **220** may be less than or equal to 2 mm. Further, the distance L**1** between the display screen assembly **210** and the protective member **220** may be less than or equal to 1 mm. For example, the distance L**1** between the display screen assembly **210** and the protective member **220** may be 0 mm or 0.5 mm.
[0062] The smaller the distance L**1** between the display screen assembly **210** and the protective member **220**, the smaller the distance between the display area **211** and the protective member **220**, and the clearer the display effect of the display area **211** to the outside. By setting the distance L**1** between the display screen assembly **210** and the protective member **220** to less than or equal to 2 mm, it is beneficial to improve the display effect of the display area **211** to the outside.

[0063] As shown in FIGS. **4**, **9** and **11**, the converter **1** may have a first direction D**1** and a second direction D**2** that intersect each other. The converter **1** further includes a light source assembly **300**, a direction in which a light emitting surface **301** of the light source assembly **300** faces is the first direction D**1**, and a direction in which the display area **211** faces is the second direction D**2**.
[0064] When a user looks at the display area **211** horizontally, the display area **211** can face the user in the second direction D**2**, which is beneficial for the user to clearly observe the display area **211**.
[0065] The light source assembly **300** can indicate a working state of the converter **1** by blinking, changing colors, or the like. When the user looks at the display area **211** horizontally, the light emitting surface **301** emits light in the first direction D**1**, so that a light emitting direction of the light emitting surface **301** is not toward the user's eyes, thereby not dazzling the user, reducing discomfort of the user's eyes, and facilitating the user to clearly observe the display area **211**.
[0066] As shown in FIGS. **4**, **9** and **11**, the outer side of the converter body **100** is further provided with a reflective groove **112**. The mounting groove **111** and the reflective groove **112** are located on the same side of the converter body **100**, and are arranged and communicate in the first direction D**1**. The reflective groove **112** has a reflective surface **113** facing the outside, and the light source assembly **300** is provided in the mounting groove **111**, and the light emitting surface **301** faces the reflective surface **113**. In the first direction D**1**, the reflective surface **113** extends obliquely from a bottom of the reflective groove **112** to an opening of the reflective groove **112**.
[0067] The light emitted by the light source assembly **300** may enter the reflective groove **112**, then shine on the reflective surface **113**, and then be reflected to the outside through the reflective surface **113**. The user can look directly at the reflective surface **113** and observe a light emitting situation of the light source assembly **300** through the reflective surface **113**. In this way, the user can clearly observe the light emitting situation of the light source assembly **300** without looking directly at the light source assembly **300**.
[0068] Further, the light source assembly **300** can be shielded by the protective member **220**, so that when the user looks at the display area **211** horizontally, the light emitting surface **301** cannot be seen.
[0069] A front the display area **211** faces along the first direction D**1** may be an observation position, and the reflective surface **113** faces the observation position, so that the user can observe the display information on the display area **211** and the light emitting situation of the light source assembly **300** at the same time.
[0070] As shown in FIG. **9**, the light source assembly **300** may include a mounting frame **310** and a tape light **320** which is provided inside the mounting frame **310**, the light emitting surface **301** is located at a side of the tape light **320** facing the reflective surface **113**, and a mounting part **311** protrudes from an outer surface of the mounting frame **310** and is detachably connected with the converter body **100** in the mounting groove **111**.
[0071] Specifically, the tape light **320** can be provided in a strip shape, and the light emitting surface **301** extends along a length direction of the tape light **320**, so that the area of the light emitting surface **301** can be increased. The tape light **320** may include a plurality of light beads, which contributes to enhancing diversity of luminous intensities and luminous forms of the tape light **320**.
[0072] In the assembly process, first the light source assembly **300** and then the protective member **220** can be assembled on the converter body **100**, so that the protective member **220** can protect the light source assembly **300**.
[0073] Further, the surface cover **110** is provided with a third threaded hole **115**, and the mounting part **311** is provided with a third connecting hole **312** therethrough. The screw passes through the third connecting hole **312** and extends into the third threaded hole **115**, so that the light source assembly **300** is connected with the surface cover **110**, thus facilitating the detachment of the light source assembly **300**.
[0074] There may be a plurality of mounting parts **311** which are arranged at intervals along the

length direction of the tape light **320**. The third threaded holes **115** correspond to the mounting parts **311**, respectively.

[0075] As shown in FIGS. **2**, **3** and **11**, the converter **1** further includes a radar assembly **400** (e.g., radar circuitry) provided in the mounting groove **111**. The radar assembly **400** may include one or more antennas, duplexers, receivers, control circuit, and/or transmitters. The radar assembly **400** is electrically connected with the display screen assembly **210**, and is configured to receive and/or send a signal for controlling on or off of the display screen assembly **210**.

[0076] The radar assembly **400** can receive signals by radar sensing, and then trigger operation or shutdown of the display screen assembly **210**. Different from a conventional key switch, by providing the radar assembly **400**, a control mode of the display module **200** can be made more intelligent and flexible.

[0077] The radar assembly **400** can be electrically connected with the light source assembly **300**, receive signals by radar sensing, and then trigger operation or shutdown of the light source assembly **300**.

[0078] The present disclosure can shield assemblies or structures that do not need to be displayed inside the mounting groove **111** to beautify an appearance of the converter **1**. By setting the translucent area **221** to be translucent, the translucent area **221** can allow the lighted characters and images on the display area **211** to be seen by users, and the translucent area **221** can also shield the unlighted characters and images on the display area **211**, thus limiting visual interference caused by the unlighted characters and images and making the lighted characters and images stand out, thus improving the display effect of the display area **211** and facilitating the users to clearly observe the display information on the display area **211**.

[0079] The above are only examples of the present disclosure and do not limit the scope of the disclosure. Any equivalent structure or process transformation made using the contents of description and drawings of the present disclosure, or direct or indirect use thereof in other related technical fields should be included in the scope of the present disclosure.

Claims

1. A power converter comprising: converter circuitry configured to convert electrical energy; a housing comprising a mounting groove; and a display comprising: a display screen provided in the mounting groove; and a protective layer covering the mounting groove, wherein: the protective layer has a translucent area and an opaque area adjacent to the translucent area, the display screen comprises a display area located corresponding to the translucent area, and a projection of the translucent area toward the display screen covers the display area.

2. The power converter of claim 1, wherein: the protective layer comprises a transparent substrate and a light shielding layer covering part of an inner side of the transparent substrate facing the mounting groove, and an area where the light shielding layer covers the transparent substrate forms the opaque area, the protective layer further comprises a translucent layer covering another part of the inner side of the transparent substrate facing the mounting groove, and an area where the translucent layer covers the transparent substrate forms the translucent area, and the light shielding layer is made of opaque material, and the translucent layer is made of translucent material.

3. The power converter of claim 2, wherein visible light transmittance of the translucent layer is 5% to 25%.

4. The power converter of claim 1, wherein the display screen comprises a translucent film attached to the display area, and visible light transmittance of the translucent film is 30% to 60%.

5. The power converter of claim 1, wherein an outer side of the protective layer facing away from the mounting groove is provided with at least one of an anti-fingerprint coating and an anti-glare coating.

6. The power converter of claim 1, wherein: the display further comprises a plate, and the

protective layer and the display screen are respectively provided on two opposite sides of the plate; the plate comprises a mounting hole; and the display area is exposed to the translucent area through the mounting hole.
7. The power converter of claim 6, further comprising: a surface cover, wherein: the mounting groove is formed on the surface cover, a side of the surface cover facing away from the mounting groove is connected to the housing, and the plate is detachably mounted on the surface cover.
8. The power converter of claim 1, wherein a distance between the display screen and the protective layer is less than or equal to 2 mm.
9. The power converter of claim 2, wherein: the transparent substrate is tempered glass; the opaque material comprises at least one of opaque black ink, opaque black film and black chemical plating; and the translucent material comprises at least one of translucent black ink or translucent film.
10. The power converter of claim 1, further comprising: a light source, wherein a direction in which a light emitting surface of the light source faces is a first direction, and a direction in which the display area faces is a second direction intersecting with the first direction.
11. The power converter of claim 10, wherein: the housing is further provided with a reflective groove; the mounting groove and the reflective groove are located on the same side of the power converter and are arranged and communicate in the first direction; the reflective groove has a reflective surface, the light source is provided in the mounting groove, and the light emitting surface faces the reflective surface; and the reflective surface extends, in the first direction, obliquely from a bottom of the reflective groove to an opening of the reflective groove.
12. The power converter of claim 11, wherein: the light source comprises a mounting frame and a tape light provided inside the mounting frame, the light emitting surface is located at a side of the tape light facing the reflective surface, and a mounting part protrudes from an outer surface of the mounting frame and is detachably connected with the housing in the mounting groove.
13. The power converter of claim 1, further comprising: radar circuitry configured to receive a signal for controlling on or off of the display screen.
14. A power converter comprising: converter circuitry configured to convert electrical energy; a display; and a protective layer covering the display, wherein: the protective layer comprises a transparent substrate, a light shielding layer, and a translucent layer, and the display is configured to display information indicating a state of the power converter via the transparent layer.
15. The power converter of claim 14, wherein the protective layer has a translucent area and an opaque area adjacent to the translucent area.
16. The power converter of claim 14, wherein the light shielding layer covers part of an inner side of the transparent substrate and the translucent layer covering another part of the inner side of the transparent substrate.
17. The power converter of claim 14, wherein the display comprises a display area and a translucent film attached to the display area, and visible light transmittance of the translucent film is 30% to 6**0**%.
18. The power converter of claim 14, wherein: the transparent substrate is tempered glass; the light shielding layer is made of opaque material; and the translucent layer is made of translucent material.
19. The power converter of claim 14, further comprising: a light source comprising a mounting frame and a tape light provided inside the mounting frame; and a light emitting surface located at a side of the tape light, wherein a mounting part protrudes from an outer surface of the mounting frame and is detachably connected with the power converter.
20. The power converter of claim 14, wherein visible light transmittance of the translucent layer is 5% to 25%.